Chekurthi Deepak

Java Backend Engineer

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SUMMARY

Java backend engineer with 2 years building distributed, event-driven microservices on Spring Boot, JMS, and AWS. Designed and independently shipped an automated error-recovery framework spanning 111 production services with zero changes to existing code. Additional depth in retrieval-augmented generation and applied LLM pipelines, built and measured against labelled evaluation sets.

TECHNICAL SKILLS

Languages	Java, Python, SQL, TypeScript, JavaScript
Backend	Spring Boot, REST APIs, Microservices, Event-Driven Architecture, Distributed Systems, FastAPI
Messaging	RabbitMQ, ActiveMQ Artemis, JMS, AWS SQS, Celery, WebSocket, STOMP
Databases	PostgreSQL, Microsoft SQL Server, MySQL, MongoDB, pgvector
Cloud and DevOps	AWS (EC2, S3, SQS, RDS, Lambda, CloudWatch), Docker, GitHub Actions, CI/CD
AI and Retrieval	RAG pipelines, vector search, Sentence-Transformers, CLIP, LLM APIs, retrieval evaluation
Testing and Tools	JUnit 5, pytest, distributed tracing, Git, Maven, Postman
Frontend	Angular, React

EXPERIENCE

UST HealthProof — **Aug 2024 - Present**

Developer II — *Thiruvananthapuram, India*

- Designed a transaction-completion framework in Java and Spring Boot that reliably detects when a distributed transaction is settled and broadcasts real-time completion events to every dependent microservice, removing a longstanding bottleneck in the platform's consolidation pipeline.
- Independently built an automated error-recovery system from the ground up covering 11 modules and 111 microservices with zero changes to any existing service; failures are classified and assessed for recoverability automatically with full traceability, sharply reducing manual retry operations that previously required team intervention on every failure.
- Delivered the execution layer of that recovery system, providing configurable, track-specific reprocessing strategies for failed transactions plus an on-demand REST interface for operations teams to trigger recovery outside the automated pipeline.
- Validated a major database platform migration, identifying and resolving critical integration defects in the Java and JPA persistence layer.

PROJECTS

OnCall Copilot, AI-Assisted Incident Response Platform [Live Demo] [Source] — **2026**

Java 17, Spring Boot 3, RabbitMQ, PostgreSQL, pgvector, WebSocket, Angular

- Built a three-service event-driven backend that correlates monitoring alerts into incidents by walking a service-dependency graph, with time-windowed flap deduplication, auto-reopen on recurrence, monotonic severity escalation, and a Postgres advisory lock guarding the concurrent-alert race, mutation-verified by deleting the lock and confirming the test fails; streams live updates to an Angular dashboard.
- Implemented a retrieval-augmented generation pipeline over historical postmortems that forces the model to cite its sources or report no match, measured at MRR 0.939 against a 0.894 lexical baseline on a hand-labelled set including negative cases that verify it declines rather than citing a weak match.
- Verified failure modes against live infrastructure rather than mocks, covering LLM provider fallback, dead-letter routing, and idempotent consumers, with distributed tracing and Micrometer metrics end to end; 66 automated tests, and 9 defects surfaced in a single verification pass against code marked complete.

Multimodal RAG, Document Question Answering over PDFs — **2026**

Python, FastAPI, PostgreSQL, pgvector, Celery, Redis, React, Docker, GitHub Actions

- Built a full-stack multimodal retrieval system indexing both text and embedded images from PDFs, fusing dual-modality vector search through a calibrated scoring model into page-cited answers from a vision-capable LLM.
- Designed a labelled retrieval-evaluation harness of 38 questions to tune the system quantitatively, improving delivered-answer rate from 75 percent to 81 percent over a text-only baseline and diagnosing a silent embedding-truncation bug.
- Built asynchronous ingestion with FastAPI, Celery, and Redis over a pluggable object-storage layer, with session-cookie and API-key authentication, rate limiting, and query caching, shipped with roughly 100 automated tests and a GitHub Actions CI pipeline.

EDUCATION

National Institute of Technology Calicut — **2020 - 2024**

Bachelor of Technology in Computer Science and Engineering — *Kozhikode, India*